



HYPERTENSIVE DISORDERS OF PREGNANCY

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LEARNING OBJECTIVES

- To understand the classification of hypertension in pregnancy.
- To appreciate and be able to differentiate the different risks associated with various types of hypertensive disorders in pregnancy.
- To understand the pathophysiology of pre-eclampsia.
- To be aware of the clinical presentation of pre-eclampsia and understand the principles of management
- To understand the long-term risks to both mother and baby from preeclampsia.



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INTRODUCTION

- Hypertension is common in pregnancy.
- Approximately 1 in 10 women will have one or more episodes of raised blood pressure prior to delivery
- The majority have a benign condition called gestational hypertension, which is not associated with adverse outcomes.
- about one-third of these women (3% overall) will develop pre-eclampsia.
Pre-eclampsia is a leading cause of maternal death
- The World Health Organization (WHO) estimates that globally between 50,000 and 75,000 women die of this condition each year. Furthermore, pre-eclampsia is frequently accompanied by fetal growth restriction (FGR), which is responsible for considerable perinatal morbidity and mortality.



CLASSIFICATION OF HYPERTENTION

- There is now a widely agreed classification system for hypertension in pregnancy. Simply put, pregnant women who develop or present with hypertension in pregnancy have one of three conditions:
- 1- Non-proteinuric pregnancy-induced hypertension.
- 2-Pre-eclampsia.
- 3-Chronic hypertension.
- Non-proteinuric pregnancy-induced hypertension (otherwise known as gestational hypertension) is hypertension that arises for the first time in the second half of pregnancy and in the absence of proteinuria.
- However, up to one-third of women who present with gestational hypertension will progress to pre-eclampsia



- Women who have confirmed hypertension in the first half of pregnancy most likely have chronic hypertension.
- The majority will have essential hypertension but this is a diagnosis of exclusion and **secondary causes** should be excluded.
- Chronic hypertension, of whatever type, can predispose to the later development of superimposed pre-eclampsia
- The physiological fall in blood pressure that occurs in the first trimester secondary to peripheral vasodilatation can mask chronic hypertension.
- For example, a booking blood pressure of 138/88 mmHg, while still technically within normal limits, raises the suspicion of an underlying hypertensive tendency.



DEGREES OF HYPERTENSION

- Mild: diastolic blood pressure 90–99 mmHg, systolic blood pressure 140–149 mmHg.
- Moderate: diastolic blood pressure 100–109 mmHg, systolic blood pressure 150–159 mmHg.
- Severe: diastolic blood pressure ≥ 110 mmHg, systolic blood pressure ≥ 160 mmHg



DEFINITIONS:

- ◎ **Hypertension** in pregnancy is diagnosed as;

Diastolic blood pressure (BP) more than **110** mmHg on any one occasion
or

Diastolic blood pressure (BP) more than **90** mmHg on two or more
consecutive occasion more than 4 hours apart.

- ◎ **Proteinuria** in pregnancy is defined as:

One 24-hour collection with total protein excretion more than **300 mg/24**
hours or

Two 'clean-catch-midstream' or catheter specimens of urine collected
more than 4 hours apart with **2+** or greater on reagent strip.



PRECLAMPSIA

- Incidence Pre-eclampsia complicates approximately 2–3% of pregnancies, but the incidence varies depending on the exact definition used and the population studied.
- In the most recent UK Confidential Enquiry (2010–2012) there were nine deaths due to pre-eclampsia, which represents a significant fall from 19 deaths in 2006–2008.
- **Pre-eclampsia is defined as hypertension of at least 140/90 mmHg recorded on at least two separate occasions and at least 4 hours apart and in the presence of at least 300 mg protein in a 24-hour collection of urine, arising de novo after the 20th week of pregnancy in a previously normotensive woman and resolving completely by the sixth postpartum week.**



RISK FACTORS FOR PREECLAMPSIA:

- First pregnancy.
- Multiparous with a previous history of pre-eclampsia.
- Pre-eclampsia in any previous pregnancy.
- 10 years or more since last baby.
- Age 40 years or more.
- Body mass index (BMI) of 35 or more.
- Family history of pre-eclampsia (in mother or sister).
- Booking diastolic blood pressure of 80 mmHg or more.
- Booking proteinuria (of $\geq 1+$ on more than one occasion or quantified at ≥ 0.3 g/24 h). Multiple pregnancy.



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CERTAIN UNDERLYING MEDICAL CONDITIONS

- pre-existing hypertension
- pre-existing renal disease
- pre-existing diabetes
- antiphospholipid antibodies.



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ETIOLOGY AND PATHOPHYSIOLOGY

- Pre-eclampsia only occurs in pregnancy, but has been described in pregnancies lacking a fetus (molar pregnancies), suggesting that is the presence of trophoblast tissue that provides the stimulus for the disorder..
- Many studies considered the role of a key circulating vasoactive factors including interleukins, tumour necrosis factor alpha and components of the angiotensin pathway



■ The proposed etiology is summarized as follow :

■ Genetic predisposition



■ Abnormal immunogenic response



■ Deficient trophoblast invasion



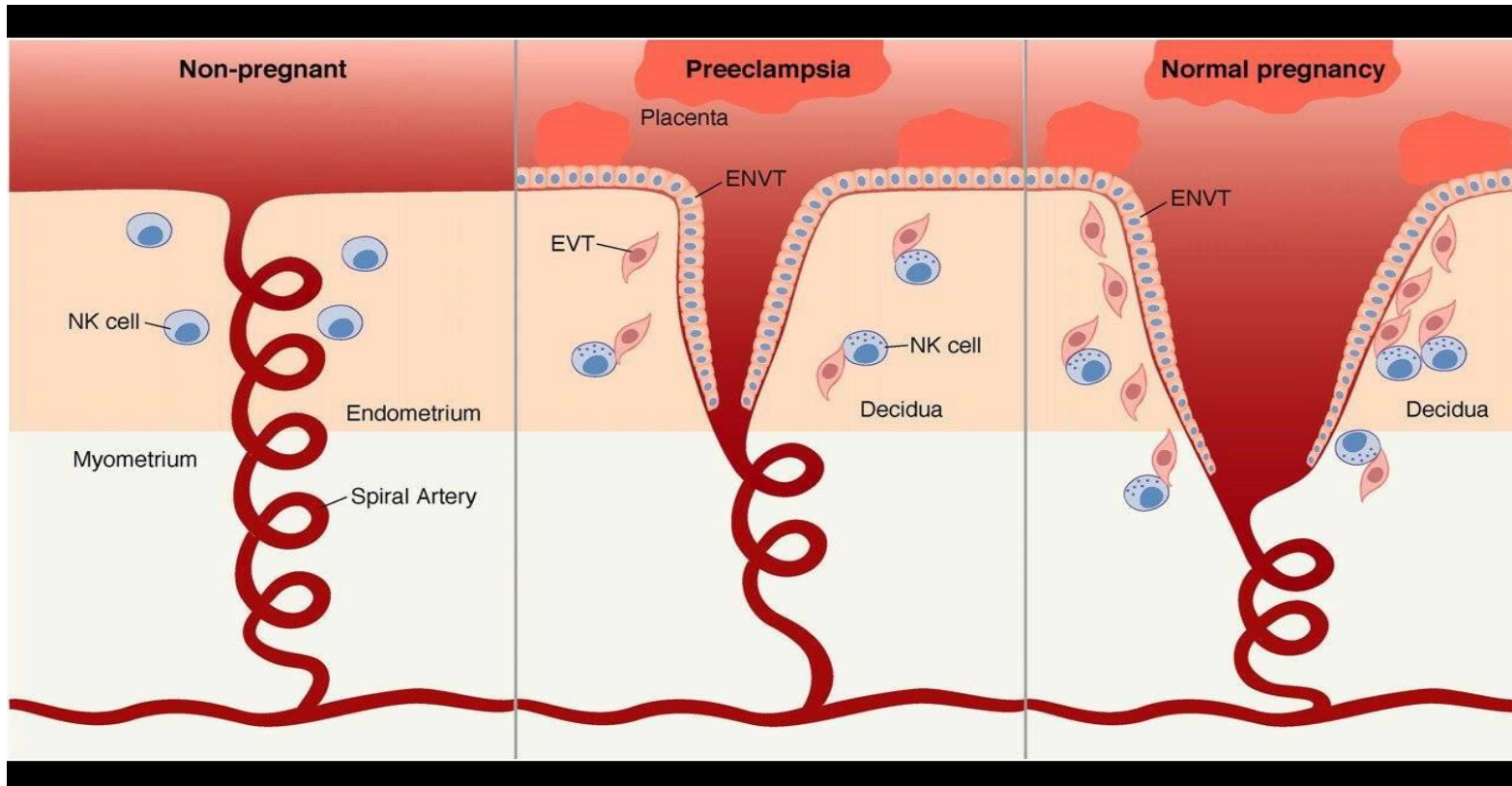
■ Hypoperfused placenta

■ Circulating factors and endothelial cell activation

■ Clinical manifestations of disease



PATHOPHYSIOLOGY ..FAILURE TROPHOBLASTIC INVASION AND SPIRAL ARTERY CONSTRICTION





ETIOLOGY AND PATHOPHYSIOLOGY

- It is widely believed that defective trophoblast invasion results in relative under-perfusion of the placenta and that this result in release of factors into the maternal circulation that targets the vascular endothelium
- pre-eclampsia is a multisystem disease it **affects** multiple organs concurrently like **cardiovascular** system resulting hypertension and edema, **renal system** resulting protienuria, **hematological** disorder resulting thrombocytopenia, the **liver** resulting elevated liver enzyme and HELLP syndrome, and **CNS** resulting hypertensive encephalopathy



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CLINICAL PRESENTATION

- The classic symptoms of pre-eclampsia include a frontal head ache, visual disturbance and epigastric pain. However, the majority of women with pre-eclampsia is asymptomatic or merely complains of general vague 'flu-like' symptoms.
- severe headache , problems with vision, such as blurring or flashing before the eyes , severe pain just below the ribs , vomiting , sudden swelling of the face, hands or feet.
- During history taking enquiry about above mentioned risk factors should be done as risk factors are found in one third of the patient.



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IMMINENT ECLAMPSIA

- cerebral signs may immediately precede the convulsion, such as nausea, vomiting, headaches, and cortical blindness.
- If the complication of multi-organ failure ensues, signs and symptoms of those failing organs will appear, such as abdominal pain, jaundice, shortness of breath, and diminished urine output



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CLINICAL PRESENTATION

- Clinical examination should include a complete obstetric and neurological examination. HT is usually the first sign, but occasionally is absent or normal or transient until the late stage of disease.



HOW TO CHECK THE BLOOD PRESSURE?

- For correct measurement the patient should be rest for more than 10 minutes or hospitalized, be in either sitting or left lateral position and the arm should be at the level of heart,
- mercury sphygmomanometer is preferred and it should be with appropriate cuff size that fit the upper arm of pregnant women (that is the bladder encircle more than 80% of arm).
- Korotkoff phase 5 is preferred over phase 4.



- Dependent edema of the feet is very common in healthy pregnancy. However, rapidly progressive edema of the face and hand may suggest pre-eclampsia.
- Epigastric tenderness is a worrying sign and suggests liver involvement.
- Neurological examination may reveal hyperreflexia and clonus in severe cases.
- Urine testing for protein should be considered part of the clinical examination. BMI should be calculated



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TESTING FOR PROTEINURIA:

1-Dipstick analysis: the results may be as

- ⦿ Trace: seldom significant.
- ⦿ 1+: possible significant proteinuria, warrants quantifying.
- ⦿ 2+ or more: probable significant proteinuria, warrants quantifying.

2-Protein: creatinin ratio:

If more than 30 mg/mmol means probable significant proteinuria

3-24 hour collection:

If more than 0.3g/24 hour represents confirmed significant proteinuria.

COMPLICATIONS:



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Maternal complications:

- CNS involvement(seizure& stroke)
- Disseminated intravascular coagulopathy.
- Renal failure.
- Hepatic failure.
- HELLP syndrome.
- High rate of C/S.
- Ante partum and postpartum hemorrhage.
- Death.
- Future risk of chronic hypertension.



Fetal complications:

- Intrauterine growth restriction.
- Placental infarct and IUD.
- Placental abruption.
- Prematurity.
- Perinatal death.
- Future cardiovascular disease.



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MANAGEMENT AND TREATMENT:

- There is no cure for pre-eclampsia other than to end the pregnancy by delivering the baby and the placenta.
- So delivery of the baby will be a good decision if pregnancy is at or very near to term as there will be no maternal or fetal reason to continue beyond that.



- This can be significant problem, however, if it occurs early in pregnancy, particularly at gestations below 34 week.
- In such cases and when the disease is sever, steroid should be administered to enhance lung maturity and delivery should be prompted.



- **indications for premature delivery are:**
- Sever uncontrolled BP,
- deteriorating liver and renal functions,
- hematological and neurological complications and
- deteriorating fetal conditions.



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- Therefore, management strategies are aimed at minimizing risk to the mother in order to continued fetal growth so if the pregnancy is at term, termination of pregnancy should be decided immediately after stabilizing the mother.



■ *The principles of management are:*

- ⦿ Early recognition of the symptomless syndrome;
- ⦿ Awareness of the seriousness of the condition;
- ⦿ Use of agreed guidelines for managements and use of antihypertensive and anticonvulsant drugs;
- ⦿ Well-timed delivery to avoid maternal and fetal complication;
- ⦿ Post-natal follow up and counseling for future pregnancy.



- ⦿ A diagnosis of pre-eclampsia usually requires admission.
- ⦿ Patient with mild hypertension, minimal proteinuria and normal hematological and biochemical test may be monitored as outpatients but will require frequent visit attendance for fetal and maternal assessment.
- ⦿ Women with moderate to severe HT, significant proteinuria or abnormal testes require admission and in patient managements. Investigation indicated in the ongoing management of pre-eclampsia are the following:



INVESTIGATION:

To monitor maternal conditions:

- ⦿ Full blood count with particular attention on falling platelet count and rising haematocrit.
- ⦿ Coagulation profile if platelet counts are abnormal.
- ⦿ Serum renal function test like urea, creatinin and uric acid.
- ⦿ Serum liver profile (SGPT,SGOT ...)
- ⦿ Urine for albumin.

To monitor fetal complication:

- ⦿ Ultrasounds for fetal size and amniotic fluid volume.
- ⦿ Antenatal cardiotocography (CTG).
- ⦿ Fetal and maternal Doppler.



MANAGEMENT REMOTE FROM TERM

- Early onset pre-eclampsia is frequently associated with placental insufficiency.
- Asymphyseal-fundal height should be carefully measured , in addition to an enquiry about fetal movement.
- Apart from ultrasound for fetal growth and Doppler study for suspected fetal compromise, involvement of other organ systems must be sought.



ANTIHYPERTENSIVE DRUGS:

the aim of antihypertensive drugs is to lower the BP (aiming at diastolic blood pressure not less than 80mmHg) without reducing uterine blood flow and compromising the fetus. There are a variety of antihypertensive drugs.

- ◉ **Methyldopa** is a centrally acting antihypertensive agent with recorded long term safety during pregnancy. Its available in oral tablet of 250 mg. however, it can only be given orally, it takes upwards of 24 hours to take effect (not used for emergency) and has a range of unpleasant side effects, including sedation and depression.



- **Labetalol** is a combined alpha and beta blocking agent, it also has long term safety profile and can be given both orally and intravenously.
- **Hydralazine** is act by vasodilatation and also can be given orally and intravenously and is used also in emergency management of hypertension.



- Both labetalol and hydralazine can be given by infusion in severe fulminating conditions.
- **Nifedipine** is a calcium-channel blocker with a rapid onset of action. It can, however, cause severe head ache that may mimic worsening disease.
- **Magnesium sulphate** (although used to control eclampsia) but can also be used in pre-eclampsia to prevent the onset of convulsion.



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LABOUR AND DELIVERY:

- The time and mode of delivery depends on many factors like gestational age, severity of the conditions, prior obstetric history, and fetal conditions.
- Iatrogenic premature delivery is often required in severe pre-eclampsia. If her condition allowed the mother should be transferred to a center with adequate facilities to care for premature baby.



- ⦿ Steroid should be given and the delivery in these cases is usually by caesarean section.
- ⦿ Such patients are at particularly high risk for thromboembolism and should be given prophylactic subcutaneous heparin and to wear antithrombotic stocking.
- ⦿ Epidural anesthesia can be used if platelet and clotting factors are normal
- ⦿ If the vaginal delivery is decided (whether spontaneous or by induction), BP should be measured frequently (every 15 minutes) and high reading during labour can be controlled by both intravenous labetalol and hydralazine.



- Appropriate fluid management is necessary to avoid fluid overload as they are prone to pulmonary edema. Most protocol prefers crystalloid.
- A Foley's catheter should be inserted to monitor input and output. Avoid use of ergometrine as it rises BP.



ANESTHESIA:

- General anesthesia poses risk, as endotracheal intubation may cause severe HT.
- regional blockade is the preferred method of analgesia for labour and of anesthesia for operative delivery provided that platelet counts are more than $80 \times 10^9/L$.
- Epidural also poses the risk of prolonging second stage



POST PARTUM CARE:

- The patient should be intensively monitored for at least 48 hours post partum as most of eclamptic fit occurs at this time.
- As blood pressure may remain high in few days post partum, antihypertensive may need to be continued.
- All patients should be reviewed 6-12 weeks post partum and in addition of blood pressure checking, the renal and liver function tests should be reviewed.
- Any predisposing factors should be investigated and implication on future pregnancy should be discussed.
- Patient should be provided with appropriate safe contraception



- Many drugs can be used postpartum, but the methyldopa should be changed before 5 days postpartum as it cause depression.
- Antihypertension is given if diastolic is 100mmHg and when started the Bp should be checked every 6 hour for 48 hours before discharg.
- In community based patient care the Bp should be reviewd in 2-5 days post partum



THANK YOU

